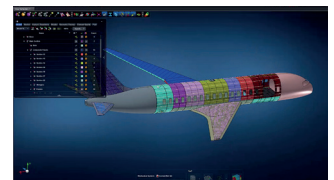




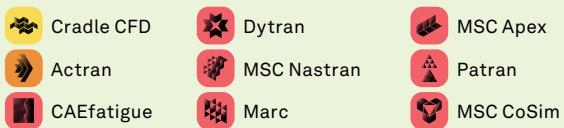
Accurately simulate real-world physics behaviour to gain valuable insights and inform innovative designs. Test and validate designs virtually to reduce physical prototypes and improve product performance with the HxGN Multiphysics suite. The HxGN Multiphysics suite accelerates productivity and streamlines workflows with advanced meshing and direct modelling capabilities, easy-to-use tools for model preparation and cleanup, high-performance computing, and specialized modules for common industry vertical workflows. **Our Multiphysics Solutions:**

Virtual prototype validation and certification

Hexagon's solvers are industry-recognized for accuracy and reliability, making it possible to validate and test designs in virtual environments without relying on cost-prohibitive physical prototypes. In some industries, like aerospace, Hexagon offers the only solvers trusted enough to virtually certify products for safety and durability.



Products

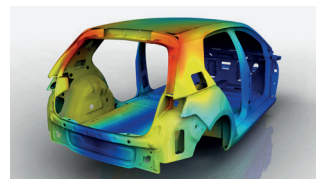


Solutions for

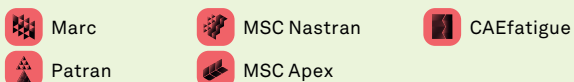
- Fatigue and durability
- Lightweighting
- Virtual prototyping
- NVH and eNVH
- Thermal analysis
- Acoustic performance
- Fluid dynamics
- Fluid-structure interactions
- Multiphysics co-simulation

Structural simulation and analysis

Perform multidisciplinary structural analysis across static, dynamic, linear and nonlinear domains with interoperable pre- and post-processors and industry-recognized solvers.



Products

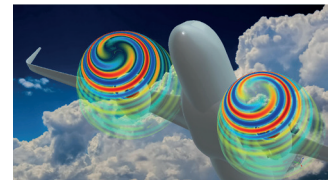
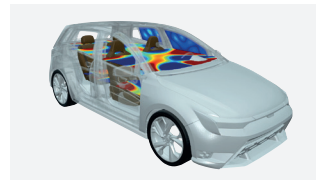


Solutions for

- Structural integrity
- Durability and fatigue
- Vibration analysis
- High-deformation scenarios

Acoustic simulation and analysis

Solve acoustics, vibro-acoustics, and aero-acoustics problems and improve the acoustic performance of designs. Integrate structural and acoustic simulation for vibroacoustic modelling to predict noise generated from vibration and vice versa.



Products

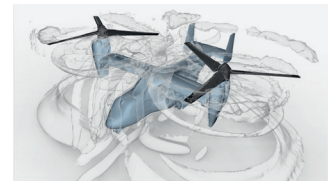
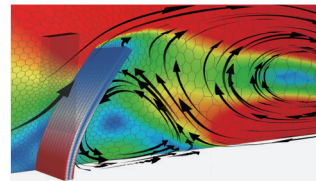


Solutions for

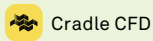
- Noise reduction and treatments
- Aeroacoustics
- High-frequency vibroacoustics
- Acoustic performance

Fluid dynamics simulation and analysis

Perform comprehensive fluid flow and thermal simulations. Manage dynamic fluid-structure interactions, capture turbulent flows, model multiphase flows, and optimise fluid machinery with high-fidelity CFD analysis capabilities.



Products

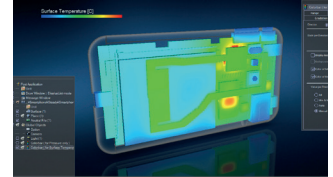
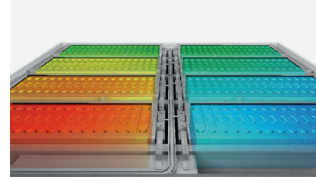


Solutions for

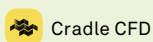
- Cavitation
- Particle tracking
- Evaporation/condensation
- Radiation

Thermal regulation and management

Process extensive simulations to solve thermal problems related to electronic devices, indoor environments, wind flow, and heat island phenomena. Simulate heat transfer effects, link structural and thermal solvers, and perform thermal stress analysis.



Products

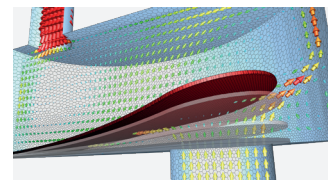
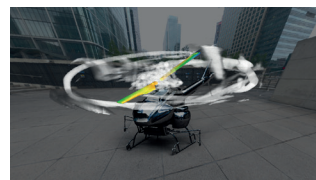


Solutions for

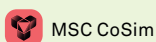
- Thermoregulation model (JOS)
- Heat conduction/transfer/thermal transport
- Thermal comfort, heat stress risk and ventilation efficiency indices

Multiphysics co-simulation

Link physics models and tools together for comprehensive virtual prototyping. For example, link CFD and FEA solvers to simulate complex fluid-structure interaction problems or explore how acoustics impact structures.



Products



Solutions for

- Fluids + multibody dynamics
- Fluids + structures
- Structures + acoustics
- Fluids + acoustics
- Many more

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